Fuel Type Analysis & US Market Entry Strategy

✅ Key Observations from Data Analysis

Based on Exploratory Data Analysis and Logistic Regression coefficients, the following variables significantly influence fuel type selection in the US automobile market:

1) Compression Ratio

Diesel vehicles are strongly associated with higher compression ratios.
The model shows a positive relationship between compression ratio and the probability of selecting diesel fuel.
This indicates that vehicles designed with high-compression engines are more likely to operate on diesel.

2) Engine Size

Diesel cars tend to have larger engine sizes compared to gas vehicles.
A larger engine size increases the likelihood of diesel selection, suggesting that diesel is preferred in vehicles built for higher load capacity or performance.

3) Horsepower

Higher horsepower is positively linked with diesel vehicles in the dataset.
This reflects customer preference for torque-oriented performance, which diesel engines typically provide.

4) Fuel Efficiency (City MPG and Highway MPG)

Diesel vehicles generally demonstrate strong fuel efficiency characteristics.
Customers choosing diesel vehicles appear to value long-term fuel economy, especially for highway driving conditions.

5) Price

Diesel vehicles are often positioned slightly higher in price compared to gas vehicles.
This suggests that diesel cars are targeted toward a more premium or performance-focused segment of customers.

🎯 Strategic Insights for Geely Auto

Insight 1: Diesel Customers Prefer Performance and Efficiency

The analysis indicates that customers who choose diesel vehicles prioritize:

- Higher engine compression
- Larger engine capacity
- Strong torque and performance
- Better long-distance fuel efficiency

This suggests that diesel vehicles are more suitable for customers seeking durability, power, and long-term operational savings.

Insight 2: Gas Vehicles Dominate the Mass Market

Gas vehicles represent the majority share of the market.
They are typically:

- Equipped with smaller engines
- More affordable
- Better suited for daily city commuting

This segment targets price-sensitive and urban customers.

Insight 3: Recommended US Market Strategy for Geely Auto

To successfully enter the US market, Geely Auto should consider the following approach:

1. **Focus primarily on gas vehicles**, as they dominate the overall market demand.
2. Introduce **select diesel models in the mid-to-premium SUV segment**, where performance and durability are valued.
3. Emphasize key selling points such as:
    - Superior fuel efficiency
    - Engine reliability and durability
    - Strong performance specifications

By aligning product design with these insights, Geely Auto can strategically position its vehicles to compete effectively with established US and European brands.